

YoungWoo Kim

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KOREATECH, Chungcheongnam-do, Republic of Korea

EDUCATION

- **KOREATECH(Korea University of Technology and Education)** 2013.03 - 2020.02
B.S. Computer engineering Chungcheongnam-do, Republic of Korea
 - GPA: 2.63/4.50
- **KOREATECH(Korea University of Technology and Education)** 2020.03 - 2022.02
M.S. Computer engineering Chungcheongnam-do, Republic of Korea
 - GPA: 4.25/4.50
 - Thesis: Real-Time Lens Distortion Algorithm on an Edge Device with GPU
- **KOREATECH(Korea University of Technology and Education)** 2022.02 - 2026.08
Ph.D. Computer engineering Chungcheongnam-do, Republic of Korea
 - GPA: 4.21/4.50
 - Dissertation: Hardware-Accelerated Ray Tracing-based acceleration for Hausdorff distance computation

RESEARCH INTEREST

- Spatial AI & Digital Twins
- High-Performance Computing (HPC)
- Physical Simulation and Proximity Queries
- Virtual and Augmented Reality (VR/AR)

WORK AND TEACHING EXPERIENCE

- **Research Assistant** Chungcheongnam-do, Republic of Korea
SPIN(Spatial Intelligence) Lab. KOREATECH 2020 Spring - 2026 Spring
- **Part-time Lecturer** Chungcheongnam-do, Republic of Korea
School of Computer Engineering, KOREATECH 2024 Spring - 2025 Fall
 - JAVA Programming (Fall 2025)
 - Multi-core programming (Spring 2024)
- **Teaching Assistant** Chungcheongnam-do, Republic of Korea
School of Computer Engineering, KOREATECH 2020 Spring - 2023 Fall
 - Multi-core programming
 - Computer Graphics
 - Microprocessors
 - Algorithms
 - C Programming

Journal Papers (SCI/SCIE)

- [J.1] YoungWoo Kim, Jaehong Lee, Duksu Kim (2025). **RT-HDIST: Ray-Tracing Core-Based Hausdorff Distance Computation**. *Computer Graphics Forum*, Vol. 44, No. 7, e70229. (Pacific Graphics 2025 Journal Track).
- [J.2] YoungWoo Kim, Sungmin Kwon, Duksu Kim (2025). **RTPD: Penetration Depth Calculation Using Hardware-Accelerated Ray-Tracing**. *The Visual Computer*, Vol. 41, No. 12, pp. 9885–9899.
- [J.3] YoungWoo Kim, Hyeon-seok Yang, Duksu Kim (2022). **Real-Time Lens Distortion Algorithm on an Edge Device with GPU**. *IEEE Access*, Vol. 10, pp. 41748–41757.
- [J.4] Jaehong Lee, You Chan No, YoungWoo Kim, Duksu Kim (2026). **A Large-Depth-Range Layer-Based Hologram Dataset for Machine Learning-Based 3D Computer-Generated Holography**. *Optics & Laser Technology*, Vol. 203, pp. 115636.

Conference Papers & Posters

- [C.1] YoungWoo Kim, Duksu Kim (2026). **RT-VIS: Hardware Ray Tracing for Visibility-based Surface Reconstruction**. In *Proceedings of the Korea Computer Graphics Society Conference*, pp. 64–65.
- [C.2] YoungWoo Kim, Duksu Kim (2023). **Penetration Depth Calculation Using Hardware Accelerated Ray Tracing Core**. In *Proceedings of the Korea Computer Graphics Society Conference*, pp. 77–78.
- [P.1] Jaehong Lee, YoungWoo Kim, Duksu Kim (2025). **Color-Corrected Full Ray-Based Computer-Generated Holography**. In *Proceedings of SIGGRAPH Asia 2025 Posters*, pp. 1–2.
- [P.2] YoungWoo Kim, Duksu Kim (2024). **Acceleration of 3D Surface Reconstruction Using Hardware-Accelerated Ray Tracing**. In *Proceedings of the Korea Computer Graphics Society Conference*, pp. 113–114.
- [P.3] YoungWoo Kim, Duksu Kim (2022). **Real-Time Lens Distortion Algorithm on Embedded GPU Systems**. In *ACM SIGGRAPH 2022 Posters*, pp. 1–2.
- [P.4] YoungWoo Kim, You Chan No, Sanghyun Lee, Duksu Kim (2022). **Simulation-Based Dataset Generation Method for Computer Vision Problems**. In *Proceedings of the Korea Computer Graphics Society Conference*, pp. 95–96.

RESEARCH GRANTS AND PROJECTS

- NRF, *Real-Time 3D Simulation via Ray-Tracing Core-Based Proximity Queries and 3D Reconstruction*, 2024–2026
- NRF, *High-Performance CGH Technology for Ultra-High-Resolution Hologram Generation*, 2021–2026
- IITP, *Holo-TV Core Technology Development for Hologram Video Services*, 2024
- IITP, *Development of Haptic Controller*, 2020–2022
- LINC+, *Development of an Autonomous Cart Platform with Scalability*, 2020–2022
- TIPA, *AI-Based Low-Cost 3D Robot Vision Platform Development (with Hana Vision Tech Co.)*, 2020–2021
- NRF, *High-Performance Scientific Visualization Algorithms on Heterogeneous Parallel Computing Environments*, 2020–2021
- KOREATECH, *Visualization of 30 Years of KOREATECH Campus Evolution Using a Shape-Changing Display*, 2020–2021
- ETRI, *High-Performance Diffraction Computation for Accelerating Ultra-High-Resolution Hologram Generation*, 2021
- ETRI, *Large-Scale Matrix Processing Module for Accelerating Hologram Generation*, 2020
- Industry-funded projects with Samsung Electronics (2023) and MarkAny (2024) (confidential)

SKILLS

- **Programming:** C++, CUDA, Java, Python
- **Spatial Computing:** 3D Reconstruction
- **GPU Computing:** CUDA, RT Cores, GLSL, Compute Shader
- **Computer Graphics:** Point Clouds, Mesh Processing, Ray Tracing
- **Development Tools:** Git, Visual Studio, Visual Studio Code, CMake

REFERENCES

1. **Prof. Duksu Kim**
Department of Computer Science and Engineering
KOREATECH (Korea University of Technology and Education)
Relationship: Ph.D./M.S. Advisor
<https://sites.google.com/view/duksukim/>